

BIOL 772 Matrix Math

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1 Basics

Start with two equations, two unknowns

$$\begin{aligned}ax + by &= u \\ cx + dy &= v\end{aligned}$$

Solve to get

$$\begin{aligned}y &= \frac{av - cu}{ad - bc} \\ x &= \frac{ud - bv}{ad - bc}\end{aligned}$$

Expand to higher order

$$\begin{aligned}ax + by + cz &= u \\ dx + ey + fz &= v \\ gx + hy + iz &= w\end{aligned}$$

Gets very hard. So convert to matrix notation:

$$\begin{pmatrix} a & b \\ c & d \end{pmatrix} \begin{pmatrix} x \\ y \end{pmatrix} = \begin{pmatrix} u \\ v \end{pmatrix} \tag{1}$$

$$\begin{pmatrix} a & b & c \\ d & e & f \\ g & h & i \end{pmatrix} \begin{pmatrix} x \\ y \\ z \end{pmatrix} = \begin{pmatrix} u \\ v \\ w \end{pmatrix} \tag{2}$$

1. A matrix is any rectangular array of numbers.
2. If the array has n rows and m columns, then it is an $n \times m$ matrix.
3. The numbers n and m are called the dimensions of the matrix.

4. We will usually denote matrices with capital letters, like A, B, etc, although we will sometimes use lower case letters for one dimensional matrices (ie: $1 \times m$ or $n \times 1$ matrices).
5. One dimensional matrices are often called vectors, as in row vector for a $n \times 1$ matrix or column vector for a $1 \times m$ matrix
6. We will use the notation A_{ij} to refer to the number in the i-th row and j-th column.
7. If $m = n$, the matrix is called square.
 - (a) A is diagonal if $a_{ij} = 0, i \neq j$.
 - (b) A is called the identity matrix

$$\begin{pmatrix} 1 & 0 & 0 & 0 \\ 0 & 1 & 0 & 0 \\ 0 & 0 & 1 & 0 \\ 0 & 0 & 0 & 1 \end{pmatrix} \quad (3)$$

- (c) The zero matrix.
- (d) A square matrix L is said to be lower triangular if $l_{ij} = 0, i < j$.
- (e) A square matrix U is said to be upper triangular if $u_{ij} = 0, i > j$.
- (f) A square matrix A is called symmetric if $a_{ij} = a_{ji}$.
8. Matrices are equal if every entry is equal
9. Matrix addition and subtraction
10. Scalar multiplication
11. Matrix multiplication

Back to biology

Given structured population and transitions between stages, you can make a basic equation $Ax = b$

Eigenvalues and eigenvectors

$$Ax = \lambda x$$

x is a special vector that is called an eigenvector, which tells me the amount of change in each dimension

λ is the eigenvalue, which tells me the amount of change

You usually have one eigenvalue/vector pair per dimension, and the largest eigenvalue in absolute value tells you what is going to happen to your system.

Solved by $\det(A - \lambda I) = 0$

We will use R to solve for us.

Need to see up system for our generic population